Validation matrix

Every reference case is an automated test (`cargo test`). A case passes only if the engine reproduces the published value within the stated tolerance. For Monte Carlo outputs the tolerance is ≈ 5 standard deviations of the engine's own dispersion, measured with `core/examples/mcm_dispersion.rs` (20 independent seeds, $M = 2 \times 10^6$), plus the rounding of the published value. Where published MCM values carry their own sampling noise, the primary reference is an analytic solution or an independent 2×10^8-trial PCG64 run (`validation/reference_values.py` → `reference_values.json`).

Test files: `core/tests/gum_examples.rs`, `core/tests/jcgm101_examples.rs`, `core/tests/eurachem_examples.rs`, `core/tests/topdown_examples.rs`, `core/tests/ea402_examples.rs`; unit tests in `core/src/*`.

Numerical building blocks

Item	Reference	Agreement	Test
erf, erfc, Φ^{-1}	SciPy	≤ 1e-12 relative	`special::tests`
Student t quantiles ν = 1…100, 2.5, 16.7	SciPy `stats.t.ppf`	≤ 1e-11 relative	`special::tests`
GUM Table G.2 ($t_{99}(16) = 2.92$ …)	GUM	2 decimals	`special::tests`
Wichmann–Hill 64-bit vs 32-bit form	JCGM 101 C.3.3.2/3	identical for 10^5 draws	`rng::tests`
t, gamma, normal generators	exact cdf/moments	statistical tests	`rng::tests`
Distribution moments (10 types)	JCGM 101 6.4	10^6 draws	`distribution::tests`
CTrap example 10.0 V ± 0.1 V → 0.060 V	JCGM 101 6.4.3	exact	`distribution::tests`
Numerical tolerance δ, examples 1–3	JCGM 101 7.9.2	exact	`mcm::tests`
Coverage interval indices	JCGM 101 7.7.2	exact	`mcm::tests`
χ^2 quantiles ν = 1…30	SciPy `stats.chi2.ppf`	≤ 1e-9 relative	`special::tests`
Rounding rules (MUPR Table 2, Nordtest round-up)	MUPR §12, Nordtest	exact strings	`report::tests`
Symbolic derivatives	central differences	≤ 1e-6 relative	`expr::tests`

Reference examples

ID	Source	Quantity	Published	Engine	Status
GUM-H1a	GUM H.1.6	u(d), ν_eff(d)	9.7 nm, 25.6	9.68 nm, 25.45 (25.6 with rounded u)	✅ (F1)

ID	Source	Quantity	Published	Engine	Status
GUM-H1b	GUM H.1, Table H.1	contributions	25, 9.7, 0, 0, 2.9, 16.6 nm	same	✅
GUM-H1c	GUM H.6, H.1.6	u_c, ν_{eff}, k_{99}, U_{99}	32 nm, 16.7, 2.92, 93 nm	31.67 nm, 16.76, 2.92, 92.5 nm	✅ (F1)
GUM-H1d	GUM H.1.7	u_c with 2nd-order terms	34 nm (terms 11.7, 1.7 nm)	33.8 nm (11.7, 1.7)	✅
S1-9.2.2	JCGM 101 Table 2	Gaussian additive	u = 2.00, [−3.92, 3.92]	agrees; GUF validated	✅
S1-9.2.3	JCGM 101 Table 3, Annex E	rectangular additive	analytic ±3.88	agrees	✅
S1-9.2.4	JCGM 101 Table 4	dominant rectangular	GUF ±19.9, MCM ±17.0; not validated (δ = 0.5)	agrees; validated only at ndig = 1	✅
S1-9.3	JCGM 101 Tables 6–7	mass calibration GUF1/GUF2/MCM	0.0539, 0.0750, 0.0754 mg	agrees; ref. 2×10^8: u = 0.075 484	✅ (F2)
S1-9.3a	JCGM 101 9.3.2.2	adaptive MCM δ/5	0.72×10^6 trials	converges, same results	✅
S1-9.4.2	JCGM 101 Table 8	comparison loss r = 0 (3 cases)	G1, G2, MCM	agrees with analytic χ^2	✅ (F2, F3)
S1-9.4.3	JCGM 101 Table 9	comparison loss r = 0.9 (3 cases)	MCM	agrees with 2×10^8 ref.	✅
S1-9.5	JCGM 101 Tables 10–11	gauge block (models 36, 37)	GUF 838, 32, [745, 931]; MCM 838, 36, [745, 932]	agrees	✅
EU-A1a	QUAM:2012 Table A1.3	Cd standard, u(V) = 0.07	1002.7 mg/L, u = 0.86 (0.9)	1002.6997, 0.8638	✅
EU-A1b	QUAM:2012 A1.4	volume sub-model	u(V) = 0.066 (0.07)	0.0665; u_c 0.835	✅ (F4)
EU-A2a	QUAM:2012 Tables A2.2–A2.3	NaOH standardisation	0.102136 mol/L, u = 0.000099, 5 contributions	identical	✅
EU-A2b	QUAM:2012 A2.4	sub-models (weighing, IUPAC molar mass, burette)	u(m) 0.13 mg, u(M) 0.0038, u(V) 0.013 mL, u_c 0.000099	0.122 mg, 0.0038, 0.0137 mL, 0.000101; MCM agrees	✅ (F10)
EU-A2c	QUAM:2012 Table A2.4	distribution choice	0.00011 / 0.000099 / 0.000085	not reproducible	⚠️ not tested (F10)
EU-A3a	QUAM:2012 Tables A3.2–A3.3	HCl titration (Kragten)	0.101387, u 0.00018, 7 contributions, (0.1014 ± 0.0004)	identical	✅

ID	Source	Quantity	Published	Engine	Status
EU-A3b	QUAM:2012 A3.1, A3.4, A3.6	summary table, triple determination, 25 °C, visual end-point	0.00016 (summary), 0.00016, 0.10149, 0.0003	0.00019 from summary inputs; others reproduced	✅ (F12)
EU-A4	QUAM:2012 Tables A4.2–A4.5, A4.6	pesticides: precision 0.27 (from raw duplicates), recovery t-test, binomial inhomogeneity (a)–(c), u_c/P 0.34, u 0.377	reproduced	✅ (F13)	
EU-A5a	QUAM:2012 Table A5.2, App. E.4	calibration from 15 raw points; u(c_0)	b_1 0.2410 (0.0050), b_0 0.0087 (0.0029), r 0.997, S 0.005486, u 0.018	identical	✅
EU-A5b	QUAM:2012 Tables A5.3–A5.4	Cd release budget and sub-models	r 0.015065, u 0.001465, u(V_L) 1.83, u(a_V) 0.19	0.001470 (exact), 1.83, u(a_V) 0.152	✅ (F14)
EU-A6	QUAM:2012 Tables A6.2–A6.5	crude fibre: s_R + drying allowance	0.31/0.4/0.6; U 0.62/0.8/0.12	U at 10 % = 1.2	✅ (F15)
EU-A7a	QUAM:2012 Tables A7.3–A7.8	lead by double IDMS, Eq. (5)	c_x 0.053738, u_c 0.00018 / 0.00041, ΣK_bias 37.6 %	reproduced	✅ (F16)
EU-A7b	QUAM:2012 Eq. (8), (12)	c_z, M(Pb)	0.092605 (0.000028), 207.2103	reproduced	✅

Top-down examples (relative values in %)

ID	Source	Quantity	Published	Engine	Status
NT-A5	Nordtest App. 5	NH_4-N duplicates: s_r < 30 µg/L; s′_r > 30 µg/L	0.44 µg/L; 3.8 %	0.441 (46 pairs); 3.82 (26 pairs)	✅ (F7)
NT-A6	Nordtest App. 6, §5.3	O_2 duplicates: pooled s, mean, s′; u(Rw)	0.0252 mg/L, 7.50, 0.34 %; 0.60 %	same	✅
NT-3	Nordtest §3.2, App. 4 (raw PT data)	NH_4-N: RMS_bias, ū(C_ref), u(bias), u_c, U	2.26, 1.52, 2.73, 3.20, 6.4 → 7 %	same	✅
NT-6.1a	Nordtest §6.1	one CRM	u(C_ref) 2.16, u(bias) 4.1 %	2.17, 4.15 % (4.1 with printed 2.16)	✅ (F5)
NT-6.1b	Nordtest §6.1	three CRMs	2.53, 1.92, 3.2 %	same	✅
NT-6.2	Nordtest §6.2	proficiency tests	1.8; 2.7/2.6; 4.6; 5.3 %	1.81; 2.60; 4.60; 5.28	✅ (F6)
NT-6.3	Nordtest §6.3	recovery	3.44, 0.76, 1.0, 3.6 %	same	✅
NT-8.2	Nordtest §8.2, App. 7 (raw data)	BOD: mean, s, s′; Examples A and B	214.8, 5.6, 2.6 %; 4.5/5.2/11 %; 1.69/3.76/4.12/4.87/10 %	same (RMS 3.77 unrounded)	✅ (F8)
NT-8.3	Nordtest §8.3	PCB Examples C and D	7.22/10.8/22 %; 7.6/8.1/11.4/23 %	same	✅ (F8)

ID	Source	Quantity	Published	Engine	Status
MUPR-E2	MUPR Example 2	mean recovery, 2 CRMs + PT	$\bar{R}$ 1.008, u 0.03705 (3 terms)	same	✅
MUPR-E3	MUPR Example 3	spiked samples with native analyte	96.2 %, 10.02 %; 1.5 % (A only)	same	✅
MUPR-B1	MUPR Example B1	nitrate by IC: $\bar{R}$, u_$\bar{R}$, Eq. 13, limits, U(I), U(II), contributions, B1.16, B1.18, reporting	97.2 %, 0.0285, 0.982 ≤ 2.093, 0.794/1.15, 0.729/1.21, 0.0676, 0.185c, 94.1/4.8/1.1 %, 82.9/9.5/2.1/5.6 %, 7.13, 0.57 ≤ 1.27, (0.261 ± 0.068), (50.0 ± 9.3)	same (0.963 and 0.0849 unrounded)	✅ (F9)
MUPR-12	MUPR §12, Table 2	rounding of reported results	(0.156 ± 0.012), (5360 ± 240), (5360 ± 260)	same	✅
MUPR-B2	MUPR B2 (raw Tables B2.1, B2.3)	pooled precision, two recoveries, compatibility, overall recovery, U, replicates	0.0285, 7.73 %, 0.048 ≤ 0.134, 99.6 %, 2.24 %, 0.169c, 7.41, 7.06	same	✅ (F17)
MUPR-B3	MUPR B3	regression precision model; U with F = 100; k = t(5)	0.0233 + 0.0128c (r 0.7266), 7.28, 0.137c	intercept 0.02324	✅ (F17)
MUPR-B4	MUPR B4	As in sediments, one CRM	4.52 %, 0.0381, 2.24 ≤ 2.26, (16.0 ± 1.9), U_tg	same	✅
MUPR-B5	MUPR B5	CRM + 2 spiked samples, matrix variability	$\bar{R}$ 0.9498, s(R) 0.0732, u 0.0800 / 0.0323, pairwise 0.275/1.32/1.50	same	✅
MUPR-B6	MUPR B6 (raw Tables B6.1–B6.3)	heterogeneity from 30 duplicates, 22 PT recoveries, χ^2, intervals	7.34 %, 0.2157, 3.00 %, 0.5828, 7.808 %, $\bar{R}$ 0.9174, u 0.01585 / 0.04872, χ^2 8.80, 0.160, 0.1157, 21.5, 29.7	same (χ^2 8.70 with weighted mean)	✅ (F17)
MUPR-C1	MUPR App. C, Example C1	allowance for an accepted recovery interval	Δa 27.87 %, u_a 8.1 %	8.05 %	✅ (F17)

Calibration examples (EA-4/02 M:2022)

ID	Example	Published	Engine	Status
EA-E1	Table E.1 coverage factors (26 values)	13.97 … 2.00	$t_{95.45}(\nu)$ agrees	✅
EA-S2	weight 10 kg	29.2 mg, U 58 mg	29.26 (29.2 from rounded contributions)	✅ (F18)
EA-S3	resistor 10 kΩ (Type A from 5 ratios)	10 000.178 Ω, 8.33 mΩ, U 17 mΩ	same	✅
EA-S4	gauge block 50 mm (second-order δα·θ)	49.999 926 mm, 34.3 nm, U 69 nm	same	✅ (F19)
EA-S5	thermocouple 1000 °C (two budgets)	0.641 °C; 36 229 µV, 25.0 µV	same (δt term 24.65)	✅ (F18)
EA-S6	power sensor (U-shaped mismatch)	0.933, 0.01623, U 0.032, ν_eff ≈ 310	same	✅ (F18)
EA-S7	step attenuator	30.043 dB, 0.0223 dB, U 0.045 dB	0.0222 dB	✅ (F18)
EA-S9	DMM 100 V, rectangular output	0.030 V, k 1.65, (0.10 ± 0.05) V	same	✅

ID	Example	Published	Engine	Status
EA-S10	vernier calliper, trapezoid β = 0.33	32 µm, k 1.83, (0.10 ± 0.06) mm	same	✅
EA-S11	temperature block, β = 0.43	164 mK, k 1.81, (180.1 ± 0.3) °C	k = 1.80	✅ (F18)
EA-S12	water meter (ν_eff = 10)	199.93 l, 0.109 l; 0.68e-3; 0.91e-3, k 2.28, U 0.002	same	✅
EA-S13	ring gauge 90 mm (temperature products)	0.15 µm sub-budget, 0.411 µm, U 0.9 µm	0.148, 0.41, 0.82 → 0.9 only with round-up	✅ (F18)

NIST Technical Note 1900 (2015) — library projects

ID	Example	Published	Engine	Status
TN-E1	weighing (linear, Gaussian)	50.284 g, u 0.0071 g, [50.270, 50.298] g	same; GUM validated	✅
TN-E2	surface temperature (Type A, ν = 21)	25.59 °C, u 0.872 °C, k 2.08, (23.8, 27.4) °C	same (GUM and MCM)	✅
TN-E3	falling-ball viscometer (skewed)	MCM 5.82, u 1.11, [4.05, 8.39] mPa s; “Gauss” 1.11	MCM same; 1st-order GUM 1.027, 2nd-order 1.10	✅ (F29)
TN-E4	Pitot tube (power law)	40.64 m/s, u 0.25 m/s, [40.17, 41.13]	57.48 m/s, u 0.348 m/s (= printed × √2)	✅ (F28)
TN-E8	M_r(CO_2), two rectangular inputs	44.0094, u 0.0007, exact [44.0080, 44.0108]	same (exact 44.008045, 44.010755)	✅ (F32)
TN-E9	cadmium standard (QUAM A1, rectangular P, triangular V)	1002.7 mg/L, u ≈ 0.9, [1001.0, 1004.4]	0.864; same interval	✅ (F30)
TN-E11	step attenuator (EA S7, arcsine δL_M, δL_D, δL_K; t_3 L_S)	30.043 dB, u 0.0224 dB, [30.006, 30.081]	same (MCM only)	✅ (F31)
TN-E13	thermal expansion, Gaussian / t_3 inputs, 95 % / 99 %	1.73e-5, u 0.18e-5 K^{-1}; (1.38, 2.07), (1.27, 2.18), (1.40, 2.05), (1.15, 2.30)	same (4 projects)	✅ (F31)

NIST Uncertainty Machine manual v1.4 (2020) and GUM H.2 — library projects

ID	Example	Published	Engine	Status
NUM-3	y = ab/c, triangular divisor	Gauss 28.8, u 8.7; MCM 32.2, sd 13, (18.2, 67)	same	✅ (F33)
NUM-8	end gauge, Gaussian inputs	Gauss u 31.7 nm; MCM sd 33.9 nm, 99 % (50 000 749, 50 000 927)	same; zero-sensitivity warning 13 %	✅
NUM-10	GUM H.2 R with rounded Table H.2 values	Gauss 0.07; MCM sd 0.0699, (127.595, 127.869) Ω	0.0700; same	✅ (F34)
NUM-11	Stefan–Boltzmann constant (CODATA 2010)	5.670 3725e-8, u 2.05e-13	same	✅
GUM-H.2	R, X, Z from the raw sets of Table H.2	0.071, 0.295, 0.236 Ω	0.0711, 0.2956, 0.2363	✅ (F34)
NUM-7, NUM-9	thermal expansion (t_3), viscosity	= TN 1900 E13, E3	see above	✅ (F29, F35)

ISO 13528:2022 Annex E — library projects

ID	Example	Published	Engine	Status
ISO13528-E.2	homogeneity and stability, As in chocolate	x̄ 0.18715, s_x 0.00398, s_w 0.00556, s_s 0.00060, 0.3σ_pt 0.00842; Δ 0.00660	same	✅
ISO13528-E.3	atrazine, 34 results, Algorithm A	x* 0.2570, s* 0.0395 (6 iterations, x* ± δ of each), u 0.0085; median 0.2620, MADe 0.0386, nIQR 0.0402	same	✅
ISO13528-E.3	median and nIQR	0.2620, 0.0402, u 0.0086	same	✅
ISO13528-E.4	mercury, reference value, 21 laboratories	D %, PA, z, z′, ζ, E_n (126 values)	same	✅ (flags: F36)
ISO13528-B.1	factors F_1, F_2	Table B.1 (g = 7 … 20)	from χ^2 and F quantiles	✅
ISO13528-E.1	23 results, 5 censored, Algorithm A	‘<’ ignored 26.01/7.23; deleted 26.81/5.29; 0.5 × limit 23.95/8.60	same; 0.5 × limit converged 23.959/8.600	✅ (F44)
ISO13528-E.7	robust mean of E.4 vs reference value	x* 0.03161, s* 0.0164 (21 results)	same	✅
ISO13528-E.5	reference value from a CRM, 20 paired samples	d 1.73, s 1.07, u(d) 0.24; x_pt 23.35, u 0.35	same	✅

ISO 21748:2017 Annex C and ISO 5725-3:2023 Annex J — library projects

ID	Example	Published	Engine	Status
ISO21748-C.1	CO in emissions, s_R = 0.28 g/km	U 0.56 g/km	same	✅
ISO21748-C.2	meat content (combined result)	95.6 %, u 1.98 %, U 4.0 %	95.6, 1.99, 4.0	✅
ISO21748-C.3	aerobic plate count, s′_R with s_l (7.3.2) + 3 %	shrimp 7.2/7.8, vegetables 8.4/8.9, flour 5.5/6.4 %	flour 6.29 %	✅ (F39)
ISO21748-C.4	crude fibre, s_R by range + drying	u 0.31, 0.41, 0.59 %; U 0.6, 0.8, 1.2 %	same	✅ (F38)
ISO5725-3-J.1	carbon in steel, 27 day-pairs	s_I(TO) 2.87 × 10^{-3} %	same	✅
ISO5725-3-J.2	vanadium, staggered-nested, levels 1–2	SS, MS, s_0^2, $s_{(1)}^2$, s_r, s_I(T), s_R	same	✅
ISO5725-3:1994-D.2	vanadium, levels 3–6 from the 1994 data (core test)	Table D.5 = J.5: s_r, s_I(T), s_R	same at levels 3–5; level 6 s_R 16.331 against 15.962 (negative $s_{(1)}^2$ kept, D41)	✅ (F40, D41)

JCGM 102:2011 — library projects and core tests

ID	Example	Published	Engine	Status
JCGM102-T1–T2	coverage factors, m = 1 … 50	k_p (Table 1), k_q (Table 2)	same	✅ (core)
JCGM102-6.5.2.3	ellipse and rectangle areas	26.6, 28.4; 11.8, 40.1 unit²	26.6, 28.4; 11.8, 40.2 (40.1 with k_q = 2.24)	✅ (F50)
JCGM102-6.5.4	coverage region for an expectation	$F_{2,10}(0.95) = 4.10$	same	✅ (core)
JCGM102-7.7	L, k_p, smallest region	L (transposed, F50); k_p 2.33 and 10.6 unit² from 1 000 draws; smallest region 9.4 unit² (10^6 draws)	k_p 2.27, 10.1 unit² (10^6 draws); smallest region 9.36 unit²	✅
JCGM102-9.2.2	additive model, Gaussian	Table 3: GUF 1.414, 0.500, 2.45, 2.24; MCM 2.45, 2.21; validated	same	✅
JCGM102-9.2.3	rectangular common effect	Table 4: MCM k_p 2.38, k_q 2.15; regions not validated	same	✅
JCGM102-9.2.4	dominant rectangular effect	Table 5: GUF 3.162, 0.900; MCM 2.28, 1.87	same	✅
JCGM102-9.3	polar co-ordinates, six cases	Tables 6, 7 (MCM, M = 10^7)	same (M = 10^6)	✅
JCGM102-9.4	resistance, reactance, impedance	Table 11: GUF, Alt. GUF, MCM (t_3)	same; MCM within the seed-to-seed spread of t_3	✅ (F51)
JCGM102-9.5.2	resistance thermometer, implicit	θ 20.023 2 °C, u 0.004 5 °C	same	✅
JCGM102-9.5.3	ten temperatures, implicit	Tables 15, 16	same	✅

ISO 5725-2:2019 — library projects and core tests

ID	Example	Published	Engine	Status
ISO5725-2-T5–T8	critical values and indicators	Table 5 (p = 8, 9; n = 2, 3), Table 6 (p = 8, 9, 15, 16, 40), Tables 7, 8 (p = 9)	D.1, D.2, D.5, D.6 within the printed digits (Table 6 single: ≤ 0.001); D.3 within ±0.003 of Table 6 for p = 6 … 40	✅ (core)
ISO5725-2-C.1	sulfur in coal, 8 labs × 4 levels, unequal n	Cochran C (C.1–C.4), Grubbs (Table C.4), $\hat{m}$, s_r, s_R (Table C.5)	same from Forms rounded to 3 decimals	✅ (F47)
ISO5725-2-C.1-REML	REML	Table C.6	same	✅
ISO5725-2-C.2	softening point of pitch, 16 labs, n = 2, missing cells	Cochran (Table C.10), Grubbs (Table C.11), $\hat{m}$, s_r, s_R (Table C.12)	same; s_R at level 4 1.9175 against 1.915	✅ (F48)
ISO5725-2-C.2-REML	REML	Table C.13	same	✅

ID	Example	Published	Engine	Status
ISO5725-2-C.3	creosote oil, outlying laboratory	Cochran 0.667, 0.636; Grubbs (Table C.17); m̂, s_r, s_R (Table C.18); relationships I–IV (Tables 1–4)	same; IV slope 0.770 against 0.772	✅ (F49)
ISO5725-2-C.3-REML	REML	Table C.19	same	✅
ISO5725-2-A.1	A_r, A_R	Table A.1 (5 entries)	same	✅ (core)

ISO 6143:2025 Annex B — library projects

Reference: B_LEAST output printed in the standard. Independent checks with ODRPACK (scipy.odr 1.18, `cov_beta`) and a Monte Carlo simulation of the whole calibration (4000 refits).

ID	Example	Published	Engine	Status
ISO6143-B.2.2	NO by UV, linear, 3 points	b, u(b), cov, S_res 0.6743, Γ 0.568, weighted deviations (Table B.4), 3 contents with u and covariances (Table B.6)	same	✅
ISO6143-B.2.3	N_2 by GC-TCD, linear, 8 points	b, u(b), cov, S_res 6.0455, Γ 1.6266, Table B.8	same	✅
ISO6143-B.2.3	quadratic	b, u(b), 3 covariances, S_res 1.3964, Γ 0.8664, Table B.9	same	✅
ISO6143-B.2.3	linear with u(x_4, x_7), u(x_5, x_8)	u(b), cov, Table B.10	same	✅
ISO6143-B.2.4	NO by NDIR, 12 points, linear (rejected)	S_res 272.6392, Γ 6.8352	same	✅
ISO6143-B.2.4	power function	b, u(b) and covariances within 0.3 % (ODRPACK 0.9 %), S_res 8.3804, Γ 1.1594, x 5.3456, u 0.014169	same	✅
ISO6143-B.2.4	exponential function	b, S_res 0.6572, Γ 0.3529, x 5.3357; u(b) 8.14 × 10^{-2}, 7.40 × 10^{-2}, 3.71 × 10^{-11}; u(x) 0.013280	b, S_res, Γ, x same; u(b) 3.18, 3.17, 1.27 × 10^{-6} and u(x) 0.014246 as ODRPACK (MC: 3.13, 3.12, 1.24 × 10^{-6}; 0.01406)	✅ (F42)
ISO6143-6.2.4	drift-test critical values	2√2 u for n = 10 (Note); Formula (12) 2√(1 + 20/n) u	2√(1 + 10/n) u, 2√(20/n) u	✅ (F41)

Palma et al. (2019), marine sediments — library projects

Two-interval top-down models (LOD … 2w_LOQ absolute, above 2w_LOQ relative) combined from the components of Table 9 (raw data only in the electronic supplement).

ID	Element	Published u_c⟨I⟩ / u′_c⟨II⟩	Engine	χ_ME⟨I⟩ / ⟨II⟩ published / engine	Status
Palma-Cr	Cr (OSPAR)	1.05 mg/kg / 10.5 %	1.049 / 10.49	60.8 / 60.9 — 60.8 / 60.8	✅
Palma-Cu	Cu (OSPAR)	0.612 / 6.13	0.6128 / 6.123	64.7 / 64.5 — 64.7 / 64.6	✅ (rounded components)
Palma-Li	Li (OSPAR)	0.127 / 12.7	0.1267 / 12.67	62.3 / 62.4 — 62.3 / 62.3	✅
Palma-Mn	Mn (OSPAR)	0.920 / 9.20	0.9198 / 9.198	42.6 / 42.6 — 42.6 / 42.6	✅
Palma-Ni	Ni (EPA 3051A)	1.25 / 8.44	1.253 / 8.442	7.9 / 7.7 — 7.9 / 7.7	✅
Palma-Pb	Pb (EPA 3051A)	1.90 / 10.1 (10.08 in Eq. 21)	1.895 / 10.08	65.2 / 57.6 — 65.2 / 57.6	✅
Palma-Zn	Zn (OSPAR)	0.365 / 9.13	0.3655 / 9.132	78.7 / 78.6 — 78.6 / 78.7	✅

Bettencourt da Silva (2026), MUPR spreadsheet — library projects

Fluoranthene in river sediments, Supplementary Material 06: 20 mean recoveries with u(R_k) from the sheet “Trueness”, two-interval precision model, s(R) added (χ^2 test failed), k = 2.093 (ν = 19). Expected values are recomputed by an independent script. The spreadsheet’s own uncertainties differ for the reasons of F72 and F73.

ID	Mean recovery	χ^2 / s(R)	Spreadsheet	Engine (0.1 / 10 mg/kg)	Status
Bet-weighted	$\bar{R}$ = 74.887 % (Eq. 16.1), u = 8.812 %	40.973 / 8.755 %	$\bar{R}$ 74.887 %, χ^2 40.973, s 8.755 %; U 0.161 / 3.857	(0.1335 ± 0.2127) / (13.353 ± 3.575)	✅ $\bar{R}$, χ^2, s; U by design (F72, F73)
Bet-arithmetic	$\bar{R}$ = 76.307 % (Eq. 8–12), u = 9.629 %	40.973 / 8.755 %	—	(0.1311 ± 0.2091) / (13.105 ± 3.724)	✅ recomputed

JCGM 106:2012 — conformity assessment (library projects and core tests)

ID	Example	Published	Engine	Status
JCGM106-7.3-E1	Zener diode, V_b ≤ −5.40 V	p_c 0.92	0.919	✅
JCGM106-7.3-E2	bursting strength, B ≥ 490 kPa	p_c 0.99	0.989	✅
JCGM106-7.4	motor oil, two-sided	p_c 0.66 (0.93 − 0.27)	0.663	✅
JCGM106-7.7.5	C_m = 1, 95 % locus	0.45 ≤ ỹ ≤ 0.55	0.4491 … 0.5509	✅ (core)
JCGM106-8.3.2	w = U = 2u (ISO 14253-1)	false acceptance ≤ 2.3 %	2.28 %	✅ (core)
JCGM106-8.3.3-E1	speed limit, u = 2 % v, 99.9 %	v_max ≈ 107 km/h	106.59	✅
JCGM106-8.3.3-E2	nandrolone, t_9, 95 %	A = 2.37 µg/L	2.367	✅
JCGM106-9.5.3	precision resistors	p_c 90 %, R_C 1 %, R_P 7 %; 83/7/9/1 per 100	90.4, 0.99, 6.90 %; 83.5/6.9/8.6/1.0	✅
JCGM106-9.5.4	ball bearings, gamma prior	4.2 % non-conforming; r ≈ 0.65, A ≈ 1.7 µm, R_P ≈ 7.5 %	4.24 %; 0.656, 1.672, 7.55 %	✅ (F45)

ID	Example	Published	Engine	Status
JCGM106-9.5.6	u_0 = T/6, w = 0, C_m = 2 and 10	R_C ≈ 0.1/0.04 %, R_P ≈ 1.5/0.07 %	0.098/0.041 %, 1.47/0.072 %	✅ (core)

Nordtest TR 569 ed. 6 (2026) — control charts (library projects and core tests)

ID	Example	Published	Engine	Status
TR569-T4	range factors, n = 2 … 4	d_2 1.128; D_WL 2.833, 3.470, 3.818; D_AL 3.686, 4.358, 4.698	d_2, d_3 by numerical integration (n = 2 … 10); D_WL 2.8334, 3.4693, 3.8184; D_AL 3.6859, 4.3577, 4.6982	✅ (core, F54)
TR569-T1	Zn, 60 raw values, X-chart	Figure 1: no value out of control	$\bar{x}$ 60.278, s 2.598; WL 55.08/65.47, AL 52.48/68.07; 3 values in the warning zone, 0 out of control; t = 0.83 against 60.0	✅
TR569-7-X1–X3	X-chart cases 1–3	54.5/63.9, 52.1/66.3; 53.3/65.1, 50.3/68.1; 54.0/66.0, 51.0/69.0 µg/L	same	✅
TR569-7-R1–R2	r%-chart cases 1–2	WL 1.0 %, AL 1.3 %; s_r = r/2.8 = 0.357 %	1.01/1.31 %; 0.357 %	✅ (F52)
TR569-E1	Ni by XRF, target from U = 4 %	s 0.0458; 4.49/4.67, 4.44/4.72	same	✅
TR569-E2	Co by XRF, target from LOQ	0.0748/0.0788, 0.0738/0.0798	same	✅
TR569-E3	NH_4-N, X- and R-chart	18.95/21.03, 18.43/21.55; s 0.496, WL 1.40, AL 1.83	same (s = 0.4954 from 0.559/1.128379)	✅
TR569-E4	Pb by ICP-MS	0.278/0.310, 0.270/0.318	same	✅
TR569-E5	As in CRM, tolerance ± 15 %	AL 15.3/20.7	same	✅
TR569-E6	total P, r%-chart	CV 1.67; WL 4.73, AL 6.16 %	same with the rounded CV (4.72/6.14 % unrounded)	✅ (F54)
TR569-E7	β-HCH, target 15 % around the reference value	11.2/20.8, 8.8/23.2	same	✅
TR569-E8	Cu by ICP-OES: preliminary and new limits, review	0.92/1.19, 0.85/1.26; F 1.563 (crit 1.67), s_p 0.07545, t 1.012 (crit 1.98); 0.88/1.21, 0.80/1.29; R-chart 0.28/0.36, F 1.037	same; F_crit 1.677 exact; s_p 0.07544; F (R) 1.0380	✅ (F54)
TR569-E9	Zn blank	−0.051/0.129, −0.096/0.174	same	✅
TR569-E10	B, target 10 % around the reference value	4/6, 3.5/6.5	same	✅
TR569-E11	micropipette, tolerance ± 0.01 mL	AL 0.99/1.01	same	✅
TR569-E12	3 × 8 replicate control results	s_r 0.15 (Eq. 10), s_Rw 0.270 (simplified), 0.272 (ANOVA; footnote: s_r 0.1541, s_I 0.2246)	0.1541, 0.2705, 0.2724 (s_r 0.1541, s_I 0.2247)	✅ (F53)
TR569-9	daily and statistical-control rules	Chapter 9	core tests: action limit, two out of three (D43), trend of seven, ten of eleven (Example 4: twelve values on one side)	✅ (core)

UKAS M3003 ed. 6 (2024) — library projects and core tests

ID	Example	Published	Engine	Status
M3003-B	t-table for p = 68.27, 95, 95.45, 99.73 %; B.10	table; ν_eff 21.1, k 2.13	same	✅ (core)
M3003-C7	k(95.45 %) for rectangular ⊛ normal, u_N/u_R = 0 … 2.5	26 values, 1.65 … 2.00	all within ±0.005	✅ (core)
M3003-C5	DVM with dominant resolution	u_c 0.305 mV, k 1.77, U 0.54 mV	same; MCM interval ±0.54 mV, validated	✅
M3003-K1	10 kΩ resistor	u_c 0.445 ppm; 10 000.1050 Ω ± 0.0089 Ω	same	✅
M3003-K2	power sensor, 18 GHz	u_c 1.69 %, U 3.39 %	same	✅
M3003-K3	30 dB attenuator	u_c 0.0245 dB, U 0.0491 dB	same (0.024 57); with E5 mismatch 0.0202 dB	✅ (F57)
M3003-K4	10 kg weight	u_c 24.56 mg, U 49.12 mg	same	✅ (F55)
M3003-K5	weighing machine, 200 g	u_c 0.150 mg, U 0.300 mg, +0.10 mg	0.153 mg, 0.306 mg, −0.10 mg	❌ printed (F56)
M3003-K6	10 mm gauge block	u_c 40.7 nm, U 81.5 nm	same; with the model's temperature terms 41.0 nm	✅ (F58)
M3003-K7	N-type thermocouple, 1000 °C	u_c(t) 0.641 °C; u_c(V) 25.9 µV, U 51.9 µV, 36 230 µV	0.641; 25.84, 51.7, 36 228.8	✅ (F59)
M3003-K8	DPI at 2 MPa	u_c 43.0 ppm, U 86.0 ppm, +17 ppm	same	✅ (F60)
M3003-K9	flowrate at standard conditions	12.53 L/s, u_c 0.275, ν_eff 130, U 0.55	same	✅
M3003-K10	sheet thickness	1.514 mm, u_c 0.009 63 mm, ν_eff 50, U 0.019 mm	same	✅
M3003-K11	PT100 probe; gas temperature	u_c 0.042 25 °C; 22.97 °C, u_c 0.086 38 °C	same; 0.086 35 °C	✅
CG18-H1	220 g / 0.1 mg balance: zero, 220 g (5d, 5e), 100 g and 220 g with buoyancy correction	u(E) 0.000 118, 0.001 971, 0.000 491, 0.000 149, 0.000 226 g; k 2.87, 2.00, 2.00, 2.25, 2.05	same; k = 2.04 exact at ν = 62 (guide uses GUM Table G.2)	✅

ID	Example	Published	Engine	Status
CG18-H2	60 kg multi-interval balance (2/5/10 g): zero, 10 kg, 60 kg (5d, 5e, option 2), adjusted 60 kg (5c)	u(E) 1.238, 1.552, 5.978, 5.948, 5.941, 5.739 g; k 2.52, 2.17, 2.05, 2.05, 2.05, 2.025	1.238, 1.552; 5.973, 5.944, 5.938, 5.725 (mpe 0.90 g for 3 × 20 kg F2); k exact 2.03, 2.01	✅ (F68)
CG18-H3	30 t weighbridge, substitution loads: zero, 10 t, 15 t, 30 t	u(E) 6.75, 7.98, 14.60, 23.85 kg; ν_eff 5, 9, 109, 783; U 18, 19, 29, 48 kg	6.75, 7.98, 14.60, 23.86; ν_eff 5, 9.8, 37, 157; U 17.9, 18.5, 30.3, 48.1	✅ u(E); ν_eff differs (F69)
CG18-H4	error function E = a_1I, 400 g balance, s_m = 0, 0.05, 0.25 mg	a_1 0.00083/0.00084; U(â) $5.109\cdot10^{-8}$, $5.637\cdot10^{-8}$, $1.745\cdot10^{-7}$; χ^2 12.5, 7.3; u(E_appr) Tables H4.6, H4.8; 2 points fail C2.2-2b at s_m = 0.05	same; minimal s_m 0.044 mg (C2.2-2a), 0.22 mg (C2.2-2b)	✅ (F70)
CG18-H1.4/A	weighing result in use, 220 g balance (5e fit, built-in adjustment, ΔT = 3 K)	a_1 $6.709\cdot10^{-6}$, u(a_1) $1.242\cdot10^{-6}$; α^2 $1.467\cdot10^{-8}$ g², β^2 $8.390\cdot10^{-12}$; U_gl = $2.422\cdot10^{-4}$ g + $1.150\cdot10^{-5}$R; R_min,SF 0.0729 g (1 %, SF 3)	$6.710\cdot10^{-6}$, $1.242\cdot10^{-6}$; $1.466\cdot10^{-8}$, $8.389\cdot10^{-12}$; $2.422\cdot10^{-4}$ + $1.150\cdot10^{-5}$R; 0.07292 g	✅
CG18-H1.4/B	same, adjusted before calibration (5c fit)	a_1 $3.895\cdot10^{-7}$ (sign omitted), u(a_1) $6.337\cdot10^{-7}$; tare $5.774\cdot10^{-7}$; β^2 $6.433\cdot10^{-12}$; U_gl slope $4.479\cdot10^{-6}$; R_min,SF 0.0727 g	$-3.898\cdot10^{-7}$, $6.332\cdot10^{-7}$; tare $1.155\cdot10^{-6}$ (7.4.4-5); $7.432\cdot10^{-12}$; $4.851\cdot10^{-6}$; 0.07277 g	✅ except tare (F76)
CG18-H2.4/A	weighing result in use, 60 kg multi-interval (5d fit), three PWI	a_1 $1.717\cdot10^{-4}$ (sign omitted), u $6.459\cdot10^{-5}$; α^2 1.867/9.917/16.167 g²; β^2 $4.589\cdot10^{-8}$; slopes $2.574/3.434/3.923\cdot10^{-4}$; U_gl 2.733 + $4.291\cdot10^{-4}$R, 10.190 + $5.151\cdot10^{-4}$(R − 12 kg), 20.311 + $5.641\cdot10^{-4}$(R − 30 kg); R_min,SF 598 g	same; U(W) at 12 and 30 kg 8.129 and 15.16 g (printed 10.190, 20.311)	✅ except two intercepts (F75)
CG18-H2.4/B	same, adjusted before calibration (5c fit)	a_1 0, u $6.043\cdot10^{-5}$; α^2 1.467/7.417/13.667 g²; β^2 $2.449\cdot10^{-8}$; U_gl 2.422 + $1.706\cdot10^{-4}$R, 6.616 + $2.355\cdot10^{-4}$(R − 12 kg), 11.951 + $2.744\cdot10^{-4}$(R − 30 kg); R_min,SF 502 g	same; 501.5 g	✅
GUM5-5	GUM-5 Example 5, GHG inventory (agriculture A4ci, UK 2018), independent and shared activity data	E 4282.04 kt, u 835.16 kt; Table 5.3 contributions	E 4282.14 (F_1 rounded), u 835.16; contributions to 10^{-4} kt; u_cor − u_ind as (5.9)	✅ (F84 class a/b)
GUM5-7	JCGM GUM-5:2026 Example 7, mass calibration (= JCGM 101 9.3)	GUF1 1.2340, 0.0539, [1.1285, 1.3395] mg; GUF2 0.0750, [1.0870, 1.3810]; MCM 0.0755	same; second-order estimate term 0	✅
GUM5-16	JCGM GUM-5:2026 Example 16, comparison loss, r = 0 and 0.9	Table 16.1 GUF2 δy 50/150/2550, u 50/112/502; Table 16.2 analytic u 67/121/505 (×10^{-6})	δy with the Amd.1 term 50/150/2550; u 50/111.8/502.5; correlated GUF2 (D57) 67.27/120.52/504.50	✅ (D56, D57)
GUM5-1	GUM-5 Example 1, pH by interpolation (25 °C; correlated standards; 23.7 °C)	7.0002, u 0.0041; u 0.0051; 7.0043 (printed 7.0109), u 0.0041	7.0002, 0.0041; 0.0051; 7.0043, u 0.0032 independent / 0.0041 same-standard correlated	✅ except F78
GUM5-2	GUM-5 Example 2, benzo[a]pyrene by internal standard (correlated pairs)	0.014 ng, u^2 $4.1\cdot10^{-6}$ ng², ν_eff 5.2 (G.2b), 4.8 (eq. 2.3); MCM [≈ 0, 0.032] ng	0.01403, $4.08\cdot10^{-6}$, 5.20, 4.81; MCM script [$1\cdot10^{-7}$, 0.0321]; ensemble ν_eff 2.11	✅ except F79

ID	Example	Published	Engine	Status
GUM5-3	GUM-5 Example 3, molar mass of glucose	180.1557, u 0.0038	same	✅
GUM5-4	GUM-5 Example 4, gravimetric CO_2/N_2 mixture (ISO 6142-1), 8 outputs	Table 4.6: Ar 4.0 ± 2.4 … CO_2 200 345 ± 5, H_2O 0.379, N_2 799 649 ± 5; r(CO_2, N_2) −0.86, r(Ar, N_2) −0.50; R_M (4.3.4.6)	same except CO_2 ± 3.96, H_2O 0.389, r −0.85 and −0.55; R_M identical	✅ except F82
GUM5-9	GUM-5 Example 9, torque slope by OLS/WLS (GUF) and Bayes	OLS 1.0107 ± 0.0015 [1.0055, 1.0158], ν 2.7; WLS 1.0085 ± 0.0008 [1.0069, 1.0101], ν 14; NI-Bayes 1.0092 ± 0.0011; Inf-Bayes 1.0091 ± 0.0009	same; WLS ν 13.31; Bayes by quadrature (script)	✅ except F83
GUM5-10	GUM-5 Example 10, conformity of TSPM with a lognormal prior	R_C 0.5 %, R_P 0.9 % (A_U = T_U); 0.002 %, 5.8 % (0.17); 1.7 %, < 0.1 % (0.22); specific R_P ≈ 11 % at 0.225	printed form: 0.498/0.909, 0.0020/5.78, 1.69/0.115 %; conditional h: 0.579/0.739, 0.0103/5.37, 1.80/0.055 %; posterior 11.33 %	✅ except F80
GUM5-11	GUM-5 Example 11, tumour area from boundary pixels (Fourier series, MCM)	Table 11.1 areas; MCM n = 6 65.4 ± 1.6 (11.6.1: 65.0); n = 5 65.3, n = 7 65.0; AIC/AICc/BIC minimal at n = 2	data confirmed by EMUE E5.1.5; areas ±0.1 (EMUE Table E5.1.1 likewise); MCM 65.45 ± 1.55 (EMUE 65.5 ± 1.6); IC with $\chi^2 = S/u_0^2$: n = 6/4/4; unweighted S gives n = 2	script only; F85 (+ addendum)
GUM5-12	GUM-5 Example 12, between-bottle homogeneity, Bayesian τ	MS $3.25\cdot10^{-7}/4.90\cdot10^{-7}$; τ 0.00014 ± 0.00011 [0.00001, 0.00040]; σ 0.00069; HPD τ/μ [0, 0.0008], σ/μ [0.0013, 0.0020]; median 1.1, mode 0.3 µmol/mol	same; τ $1.356\cdot10^{-4}$ ± $1.07\cdot10^{-4}$ [$5.2\cdot10^{-6}$, $3.97\cdot10^{-4}$]; median 1.13; mode 0; u*_bb 1.48 µmol/mol	✅ except F81
GUM5-14	GUM-5 Example 14, ^{207}Bi decay correction	10.810 kBq, u 0.054 kBq; Table 14.1	same	✅
GUM5-15	GUM-5 Example 15, wire rope breaking force	10 005.7 kN, u 9.9, U 19.4 kN; MCM u 10.1 kN	GUF same; MCM 9.93 kN	✅ GUF; MCM differs (F77)
CG19-E1	1000 mL flask (Tanaka density)	V_{20} 999.628 mL, u 0.017 mL, ν_eff 19, k 2.15, U 0.036 mL	999.622; 0.0169; 19.5; 2.14; 0.036	✅ (F62)
CG19-E2	100 mL pycnometer	100.014 mL, u 0.89 µL, ν_eff 13, k 2.21, U 2.0 µL	100.0142; 0.90 µL; 12.9; 2.23 (G.6.4 truncation); 2.0 µL	✅ (F62)
M3003-L3	DMM, 1 V range	0.95 V: u_c 5.37 µV; 500 mV: U 6.2 µV; 1 V: k 1.898	same	✅ (F61)

Eurachem UfS:2019 — uncertainty from sampling (library projects and core tests)

ID	Example	Published	Engine	Status
UFS-A1	nitrate in lettuce, 8 targets, classical ANOVA	x̄ 4345.5625; s_T 556.2804, s_samp 518.16089, s_anal 148.18063; 51.58/44.76/3.66 %	same (s_samp 518.16087)	✅
UFS-A1R	same, robust ANOVA (RANOVA)	4408.3237; 565.40, 319.05, 167.94	robust mean by H15 on target means 4408.319; s not reproduced	❌ method unpublished (F63)

ID	Example	Published	Engine	Status
UFS-A2	lead in soil, 10 targets, classical and ln x	317.8; 197.55, 135.43, 17.99; U′ 85.23/11.32/85.98 %; s_G 0.66775, 0.4784, 0.0567, 0.4817; F_U 2.6032, 1.12, 2.6207; u′ 0.5111	same	✅
UFS-A3	dissolved iron, 6 wells, range method	errata: s_A 0.015, s_S 0.090, s_T 0.601 mg/L; U 1.8/10.5/70 %	same	✅ (F64)
UFS-A3A	same, classical ANOVA	U 1.6/9.6/70 %	same	✅
UFS-A4	vitamin A, 40 g test samples	SS 16595/14231; s_anal 28.805, s_samp 17.224; RSD_B 6.1 % (errata)	same	✅ (F65)
UFS-A4S	vitamin A, 4 g test samples	V_samp −2662.15 → 0; RSD_anal 36.68 %	same	✅
UFS-A4Q	duplicate control chart	CL 11, WL 27, AL 36 %; all D % below WL	10.88, 27.33, 35.56 %; 0 outside WL	✅ (F65)
UFS-A5	enzyme in feed, Gy	c 2160; C 540, 270; s_r 0.033, 0.13; s_t 0.143; U′ 26.8/28.6 %	2158.5; 539.6, 269.8; 0.0325, 0.1296; 0.1427; 26.7/28.5 %	✅ (F66)
UFS-A6	Cd and P in soil, modelling (GUM)	0.32 ± 0.06, 116 ± 26 mg/kg	0.319 ± 0.059, 115.8 ± 26.1	✅ (F67)
UFS-D2	unbalanced nested design	method of moments	core test: EMS coefficients and MS_anal	✅ (core)

Uncorrected bias (Magnusson & Ellison 2008) — core tests

ID	Claim	Published	Engine	Status
ME2008-T3	formulae of Table 3	RSSu, RSSU, SUMU, SUMU_MAX, Ue	same (SUMU signs per Phillips et al. 1997, F43)	✅
ME2008-Ue	Ue ≈ 1.65u_c + |Δ| for |Δ| > 0.8u_c	stated	within 3 % for |Δ|/u_c = 0.8 … 4	✅
ME2008-F1	coverage, u(y) = 1, u_Δ = 0.5	SUMU ≈ 95 % → 97.5 % at 3u_c; Ue ≈ 98 % → 95 %; RSSu, SUMU_MAX > 95 %	95.2/97.5 %; 98.1/95.0 %; > 95 %	✅
ME2008-F2	interval widths, u_Δ = 0.3	RSSU < Ue ≈ RSSu < SUMU_MAX for |b| ≤ u(y); RSSu largest at 2u(y)	same	✅

Measured MCM dispersion (σ over 20 seeds, M = 2 × 10⁶)

Case	y	u	shortest lo	shortest hi	shortest length	symmetric lo	symmetric hi
mass calibration [mg]	4.0e-5	3.7e-5	7.4e-4	7.4e-4	1.8e-4	1.0e-4	1.2e-4

Case	y	u	shortest lo	shortest hi	shortest length	symmetric lo	symmetric hi
loss $x_1 = 0$, r = 0 [10^{-6}]	0.04	0.04	6e-5	0.16	0.16	0.006	0.23
loss $x_1 = 0.01$, r = 0	0.10	0.07	4e-4	0.29	0.29	0.03	0.30
loss $x_1 = 0.05$, r = 0	0.48	0.20	3.2	3.3	1.0	0.78	0.85
loss $x_1 = 0$, r = 0.9	0.04	0.08	2e-5	0.20	0.20	0.003	0.30
loss $x_1 = 0.01$, r = 0.9	0.10	0.12	0.34	0.50	0.34	0.03	0.40
loss $x_1 = 0.05$, r = 0.9	0.48	0.23	5.0	4.9	1.0	0.77	0.97

Example library (project files)

The GUM/MCM cases above are also stored as 38 self-checking project files (49 with the eleven NIST TN 1900 projects) in `library/` (see `library/README.md`), run by `project/tests/library.rs` and `kapsam check library` through the complete path a user takes (JSON → model → GUM/MCM → comparison → TR/EN reports). Reference-value policy: D15 in the research log. EA-4/02 S13 in the library uses the raw observations (estimate 0.2323 µm, F21), whereas the core test reproduces the printed budget.

The top-down cases (Nordtest §3, §5.3, §6.1–6.3, §8 A–D; MUPR Ex. 2, 3, B1, B2, B4, B5, B6; QUAM A4, A6) are stored as 22 top-down project files that start from the raw data wherever the documents give them (duplicate pairs, control-chart results, replicate groups, PT rounds, the 22 PT results of B6). The two projects from Bettencourt da Silva (2026) start from the 20 mean recoveries of the spreadsheet (`"type": "given"`). Tolerances reflect the documents' printed rounding; where a guide computes from rounded intermediates the tolerance is widened and the reason stated in the file (MUPR B1 at 50 mg/L, B5 with t = 4.30; Nordtest §8 B, C).

Planned cases

TN 1900 observation-equation examples (regression, random effects); TN 1900 E5 (truncated Gaussian inputs); ISO/TS 28037 data sets (the GLS engine of ISO 6143 covers the method); MUPR Appendix C allowance as a project file; cross-checks against the NIST Uncertainty Machine (F29) and Suncal. | EUR-CA-B1–B3 | Eurachem/CITAC compliance guide 2021, Annex B Examples 1–3 and Table 1 | B1 16.2–17.8 % Ni, non-compliant; B2 204.1 ng/g, compliant; B3 3.6 (lognormal) / 3.2 (normal) ng/g; Table 1 51/149, 61/164, 18/182, 44/227 | 16.164–17.836; 204.09; 3.557 / 3.151 (exact s_G 3.498); Table 1 with k = 1.6449 (227.6) | ✅ except F86 | | EMUE-E3.4 | EMUE E3.4, metals in soil (ISO 21748, ruggedness coefficients, zero-intercept calibration) | Tables E3.4.1–2; $u'(Pb)^2$ = 0.002 526; (27.8 ± 2.8) mg/kg; calibration u_rel 0.019 | same to rounding; RMS of $u(\hat{\delta})/x_{ref}$ 3.50 % (printed 3.3 %) → ± 2.9 | ✅ except F87 | | EMUE-E2.1 | EMUE E2.1, total risks of a four-component medication | R*_tot 0.0029, 0.0117, 0.0002; R_tot 0.0018 | 0.002 88, 0.011 72, 0.000 21; 0.001 80 (program), 0.001 82 (numpy) | ✅ | | EMUE-E2.2 | EMUE E2.2, vacuum gauge: three scenarios for a reference correction; sum model with a common temperature error | Tables E2.2.2–E2.2.7 (30 p_c values); u(Δp) 1.14/0.91 Pa, p_c 0.67/0.75 | all to 0.1 % (two rounding cases); 1.140/0.909, 0.669/0.745 | ✅ except F88 | | EMUE-E3.8 | EMUE E3.8, NH_3 in N_2 by permeation | y = 70.8 µmol/mol, u = 0.599 µmol/mol; E_A 59.164 kJ/mol, k_0 43.069 µg/min, $\partial\dot{m}/\partial T$ $4.81 \cdot 10^{-7}$ | 70.796, 0.598; E_A 56.64 kJ/mol, k_0 $9.1 \cdot 10^9$, $\partial\dot{m}/\partial T$ 0.82 µg min^{-1} K^{-1} | ✅ result; F89 | | EMUE-E3.3 | EMUE E3.3, NO_2 analyser, dynamic dilution with correlated reference values | V_x (E3.3.17) diag 39.5/37.0/41.2, off 1.7/2.2/3.0; A 88.94, B 4.19, V_{AB} (E3.3.18); validation 982.0 ± 4.7 (4.5) | diag 1.34/2.53/4.48, off

1.76/2.32/3.19; V_AB exactly (program); A 89.21 (88.94 by full WTLS); 981.95 ± 4.52 (4.35) | ✅ V_AB; F90 | | EMUE-E6.5 | EMUE E6.5, ephedrine in urine (WADA top-down budget and decision limit) | u_c 3.8 %; DL 11 µg/mL; Table E6.5.2 | u_c 3.89 %; DL 10.822 (rounded up to 11); means 11.83/11.67/11.47, z −0.93/−1.27/0.73 (σ = 0.5) | ✅ except F91 | | IUPAC24-4A | IUPAC 2024 Example 4A, water hardness | u ≈ 3 mg/L | H 96.65, u 3.18 | ✅ | | IUPAC24-4B | IUPAC 2024 Example 4B, silicon isotopic abundances (3 outputs) | 0.923 3873(36), 0.046 2300(22), 0.030 3827(30) | same | ✅ | | IUPAC24-4H | IUPAC 2024 Example 4H, cadmium standard by MCM with beta purity and triangular volume | γ 1.0027, u 0.0009, U95 0.0017 mg/mL | GUM 1.002 70 (0.000 864); MCM 1.002 70 (0.000 864), (1.001 03, 1.004 37) | ✅ | | IUPAC24-5C | IUPAC 2024 Example 5C, benzoic acid purity as beta(88 873.7, 7.11) | — (distribution given) | 0.999 920 (0.000 030); 95 % (0.999 851, 0.999 968) | ✅ | | IUPAC24-4I | IUPAC 2024 Example 4I, argentometric titration, conventional and K_sp models | 10.000 ± 0.026 mmol/L; K_sp model differs by 0.24 % | 10.000 ± 0.0255; 9.976 (mass balance) | ✅ except F92 | | IUPAC24-3E-4M | IUPAC 2024 Examples 3E, 4M, 4N, molar mass of iridium: Q test, Bayesian random-effects model (MCMC in the guide) | p = 0.0082; M = 192.2171(4) g/mol, median τ 0.0005 g/mol | p = 0.0084; 192.217 119 (0.000 440), median τ 0.000 506 (exact grid) | ✅ except F94 | | IUPAC24-4F | IUPAC 2024 Example 4F, CCQM-K145 Ni in bovine liver, maximum likelihood with dark uncertainty | $\hat{\mu}$ 2.043(15), $\hat{\tau}$ 0.044(14) mg/kg | 2.042 94 (0.015 44), 0.043 54 (0.014 24) | ✅ | | KOEPKE17-G | Koepke et al. 2017 Tables 1–2, Newtonian constant of gravitation (DL, bootstrap, Linear Pool, ML) | DL 6.673 80, 0.000 28, (6.673 20, 6.674 40); LP 6.673 67, 0.001 24, (6.671 18, 6.675 79); ML 6.673 78, 0.000 30 | DL 6.673 790, 0.000 279, (6.673 23, 6.674 35); LP 6.673 667, 0.001 243, (6.671 17, 6.675 79); ML 6.673 778, 0.000 300 | ✅ except F93 | | KOEPKE17-BAYES | Koepke et al. 2017 Tables 2 and 4, hierarchical Bayesian rows (NICOB §A.3, σ_j = u_j) | G 6.673 77, 0.000 33, (6.673 10, 6.674 42); ^{137}Cs 29.94 (0.19) a; ^{90}Sr 28.71 (0.16) a | 6.673 773, 0.000 329, (6.673 105, 6.674 409); 29.944 (0.185) a; 28.708 (0.160) a | ✅ except F95 | | PM22-HUBBLE | Possolo & Meija 2022, Hubble–Lemaître constant, Bayesian consensus with Laplace effects (Stan) | 71.69, 0.66, median τ 2.03 (km/s)/Mpc | 71.6915, 0.6603, 2.0336 (grid) | ✅ | | PM22-AG | Possolo & Meija 2022, plasma anion gap | 25, u 1.85 mmol/L | 25, 1.851 | ✅ | | PM22-TANK | Possolo & Meija 2022, storage tank V = πR²H | 7204 ± 54 m³ (Gauss and MCM) | 7204.3, 54.0 (both) | ✅ | | PM22-WB | Possolo & Meija 2022, Wheatstone bridge | 957 Ω, u ≈ 7 Ω; E 52 %, F 48 %, G 0.6 %, H 0.004 % | 957.2, 6.98; same shares | ✅ | | PM22-PB | Possolo & Meija 2022, atomic weight of lead in HIPB-1 (4 amount fractions and A_r) | u(x) 6, 37, 19, 45 × 10⁻⁶; r −0.02, +0.13, +0.18, −0.17, −0.90, −0.58; A_r 207.1791, U95 0.0002 | same u(x); r −0.011, 0.124, 0.177, −0.179, −0.895, −0.576; 207.17915, U95 0.000 17 | ✅ except F96 | | BIPM-ISRD-qNMR | BIPM ISRD-01, -02, -06, -07, -08: nine ^{1}H qNMR purity budgets | w_A and u_c of each table (e.g. ISRD-01 T4 0.9973, 7.30 × 10⁻⁴) | same w_A to rounding; u_c within 1 % except ISRD-01 T6 (4.10 vs 3.96 × 10⁻⁴) | ✅ except F97 | | PB18-CR | Possolo & Bodnar 2018 §4.4, chromium in soil (ISO Guide 35 C.1 data), ANOVA and hierarchical Bayes | REML 121.62 (0.95), τ 3.93, σ 2.87 mg/kg; MCMC 121.62 (1) mg/kg | 121.624; s_bb 3.930, s_r 2.874; Bayes grid 121.624 (0.9986) | ✅ | | MERKATAS19-G | Merkatas et al. 2019 Table 4, G from 16 results: DL (Knapp–Hartung), ML, NICOB Bayes, Linear Pool | DL 6.673 99 (0.000 25), τ 0.000 56; ML 6.673 90 (0.000 25); BG 6.673 89 (0.000 27); LOP 6.673 77 (0.001 17) | 6.673 993 (0.000 250), 0.000 556; 6.673 900 (0.000 252); 6.673 896 (0.000 271); 6.673 773 (0.001 168) | ✅ | | UME-PROC-19 | TÜBİTAK UME draft procedure, Examples 19.1 (DMM 10 V) and 19.2 (qNMR purity, illustrative) | 19.1 u_c 35.1 µV, ν_eff ≈ 4000, U 70.2 µV; 19.2 995.0 mg/g, u_c 1.567, ν_eff 121, U 3.13 mg/g | 35.09, 3965, 70.19; 995.00, 1.5669, 121.5, 3.134 | ✅ | | TUR-GB | TUR guard bands (Dobbert 2.5, RSS 1.5, RP-10 2.0) with a 95 % in-tolerance normal process | Suncal 1.7.1: GBF 0.914 186, 0.745 356, 0.75; PFA 0.6883 %, 0.5679 %, 0.3213 %; PFR 5.216 %, 17.70 %, 14.00 % | same to 2 × 10⁻⁶ | ✅ | | CG18-H3.4 | EURAMET cg-18 H3.4/A and /B, 30 t weighbridge in use (substitution loads, drift, ΔT = 40 K) | α² 62.133 / 25.333 kg²; β² 1.276 × 10⁻⁶ / 1.960 × 10⁻⁷; U = 16 kg + 1.79 × 10⁻³ R / 10.067 kg + 6.113 × 10⁻⁴ R; R_min 2169 / 1123 kg | 62.133 / 25.333; 1.276 × 10⁻⁶ / 1.967 × 10⁻⁷; 15.76 + 1.794 × 10⁻³ R / 10.066 + 6.128 × 10⁻⁴ R; 2169 / 1124 | ✅ except F98 | | KOEPKE17-CS | Koepke et al. 2017 Tables 3–4, half-life of ^{137}Cs (19 results) | DL 30.04 (0.04) a; LP 29.94 (0.93) a | 30.037 (0.045) a; 29.94 (0.93) a | ✅ | | KOEPKE17-SR | Koepke et al. 2017 Tables 3–4, half-life of ^{90}Sr (11 results) | DL 28.73 (0.11) a; LP 28.68 (0.77) a | 28.730 (0.110) a; 28.69 (0.77) a | ✅ | | NICOB-CO60 | NIST Consensus Builder manual §4.4, ^{60}Co, 19 laboratories: DL bootstrap, Linear Pool, Bayes; leave-one-out and bilateral DoE of DL, LP and Bayes (numpy port of the NICOB source, 10⁶ draws) | DL 7062, u 4, (7053, 7071) kBq; LP 7064, 29, (7012, 7127); Bayes 7062, 5, (7053, 7072); DoE: D exact (LP: within the MC error of the pool mean), U95 from the port | 7062.06, 4.33, (7053.4, 7070.7); 7063.8, 29.2, (7012.6, 7127.2); 7062.03, 4.59, (7053.1, 7071.4); U95 within 1.3 %; significant

pairs as the port | ✅ | | NICOB-PCB28 | NIST Consensus Builder manual §4.2, PCB 28, 6 laboratories with ν (one ν = 2) | DL 33.6, u 0.77, (32.0, 35.2) ng/g; LP 33.6, 1.55, (31.4, 36.3) | 33.600, 0.751, (32.06, 35.15); 33.64, 1.51, (31.44, 36.28); DoE U95 within 1.7 % of the port | ✅ except F99 |